Lecture 25
Composite H_0, errors, power

Review

Testing a mean: σ known

when to use it

σ is known; a random sample from a normal population, or n ≥ 30

test statistic

$z_{obs} = (\bar{x} - \mu_0) / (\sigma/\sqrt{n})$

if H_0 is true

z_{obs} ~ N(0, 1), standard normal

P-value

from the z-table, e.g. $P(Z \geq z_{obs} \mid H_0 \text{ true})$ if upper-tailed

Testing a mean: σ unknown

when to use it

σ is unknown, so we estimate it with the sample SD s

test statistic

$t_{obs} = (\bar{x} - \mu_0) / (s/\sqrt{n})$, df = n – 1

if H_0 is true

t_{obs} ~ t(n – 1), Student's t: heavier tails than N(0, 1)

P-value from the t-table

find t_{obs} between two values in row df; P is between their tail areas

Example: Marco's throwing distance

Marco, a high school quarterback, throws a football a mean of 40 yards, with σ = 2. His coach has him adjust his grip and records 20 throws. Distances are normal. (OpenStax, Try It 9.14)

$n = 20$; $\bar{x} = 45$ yards; $\sigma = 2$; $\mu_0 = 40$; $\alpha = 0.05$

(a) What are H_0 and H_A? Which tail? z or t?
(b) What are the standard error and the test statistic?
(c) What is the P-value? What is the decision?
(d) What is the critical value? Does z_{obs} fall in the rejection region?

(a) $H_0: \mu = 40$ $H_A: \mu > 40$, upper-tailed; σ known, so z
(b) $SE = 2 / \sqrt{20} \approx 0.4472$; $z_{obs} = (45 - 40) / 0.4472 \approx 11.18$
(c) $P(Z \geq 11.18 \mid H_0 \text{ true}) \approx 0 < 0.05$: reject H_0
(d) $Z_{0.95} = 1.645$; $z_{obs} = 11.18 \geq 1.645$, in the rejection region: reject H_0

Example: the first statistics test

Students believe the mean score on the first statistics test is 65. An instructor thinks it is higher and samples 10 students' scores, which are normal. (OpenStax, Example 9.16)

$n = 10$; $\bar{x} = 67$; $s = 3.197$; $\mu_0 = 65$; $\alpha = 0.05$

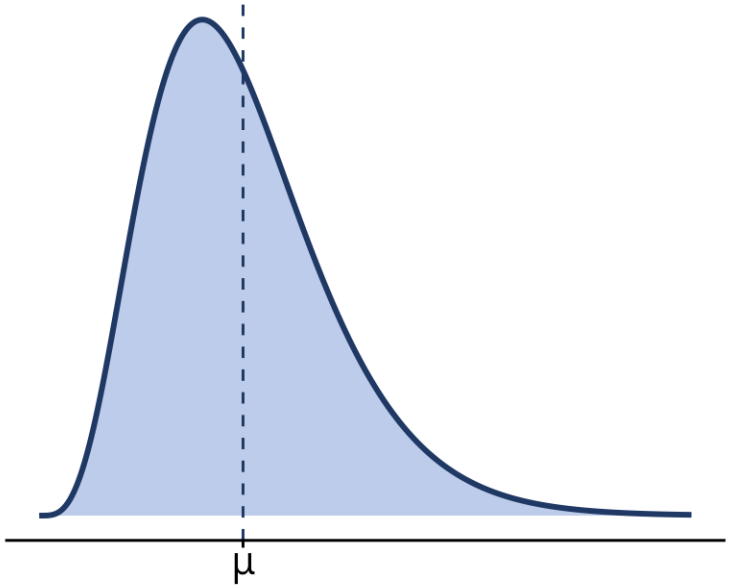
(a) What are H_0 and H_A? Which tail? Are the assumptions met?
(b) z or t? What are the standard error and the test statistic?
(c) What is the P-value (use the t-table)? What is the decision?
(d) What is the critical value? Does t_{obs} fall in the rejection region?

(a) H_0: $\mu = 65$ H_A: $\mu > 65$, upper-tailed; $n < 30$, but the scores are normal
(b) σ unknown, so t; $SE = 3.197 / \sqrt{10} \approx 1.011$; $t_{obs} = (67 - 65) / 1.011 \approx 1.98$, $df = 9$
(c) row 9: $1.833 < 1.98 < 2.262$, so $0.025 <$ P-value < 0.05: reject H_0
(d) $t_{9, 0.95} = 1.833$; $t_{obs} = 1.98 \geq 1.833$, in the rejection region: reject H_0

Three distributions: a mean

Distribution of X

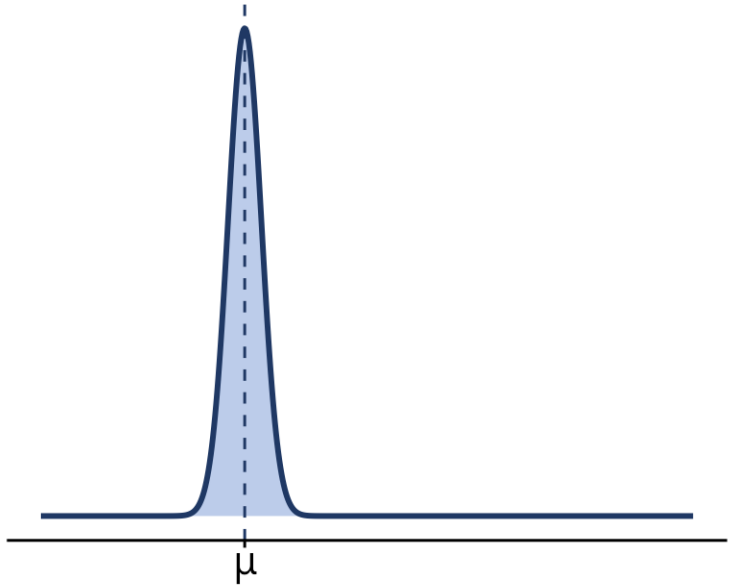
mean μ, SD σ, any shape



X, one individual

Sampling dist. of the mean

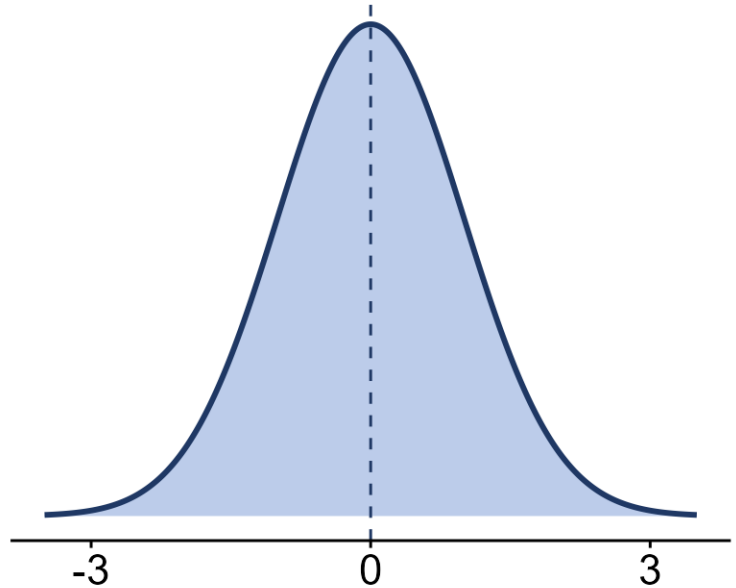
$\bar{x} \sim N(\mu, \sigma/\sqrt{n})$



x-bar, a sample of n

Distribution of z_obs

$z_{obs} \sim N(0, 1)$, if H_0 true

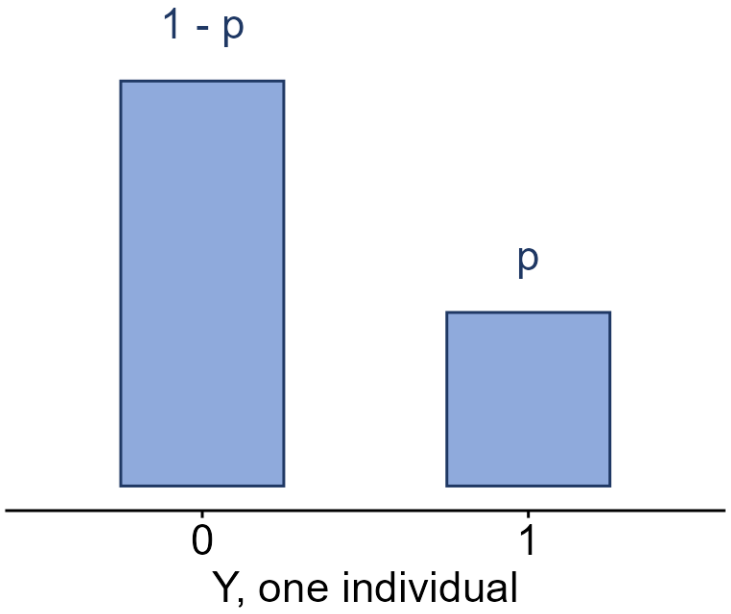


z_obs

Three distributions: a proportion

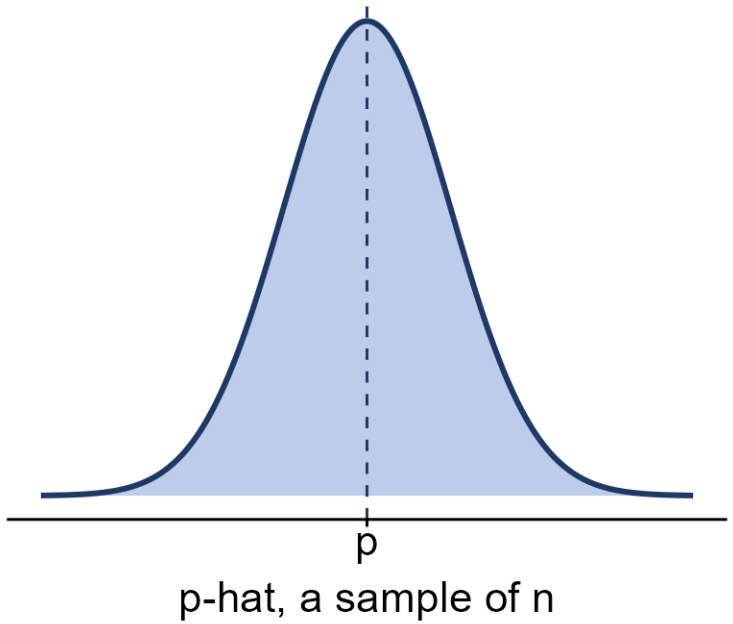
Distribution of Y

$Y \sim \text{Bernoulli}(p)$



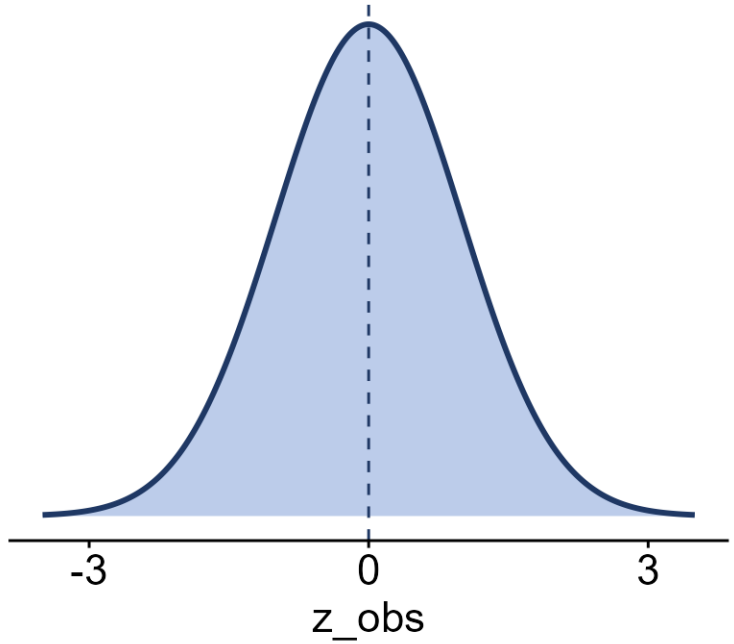
Sampling dist. of p-hat

$\hat{p} \sim N(p, \sqrt{p(1-p)/n})$



Distribution of z_obs

$z_{obs} \sim N(0, 1)$, if H_0 true



Why is z_{obs} standard normal?

a z-score

(value − mean of its distribution) / SD; ~ N(0, 1) if the value is normal

if H_0 is true

$\mu = \mu_0$, so $\bar{x} \sim N(\mu_0, \sigma/\sqrt{n})$

so z_{obs} is a z-score

$z_{obs} = (\bar{x} - \mu_0) / (\sigma/\sqrt{n})$ is the z-score of $\bar{x}$: ~ N(0, 1)

if H_0 is false

z_{obs} is centred at $(\mu - \mu_0) / (\sigma/\sqrt{n})$, away from 0

Example: the bench press

A coach's records say his players bench press a mean of 275 pounds, with σ = 55. Three players think it is more and ask 30 teammates. (OpenStax, Example 9.15)

$n = 30$; $\bar{x} = 286.17$; $\sigma = 55$; $\mu_0 = 275$; $\alpha = 0.025$

(a) If H_0: $\mu = 275$ is true, what is the distribution of $\bar{x}$?
(b) How many SDs above 275 is $\bar{x}$? What is z_{obs}?
(c) What is the P-value for H_A: $\mu > 275$? What is the decision?

(a) $\bar{x} \sim N(275, 55/\sqrt{30}) = N(275, 10.04)$, normal
(b) $(286.17 - 275) / 10.04 \approx 1.11$ SDs: $z_{obs} = 1.11$
(c) $P(Z \geq 1.11 \mid H_0 \text{ true}) = 1 - 0.8665 = 0.1335 \geq 0.025$: do not reject H_0

Composite hypotheses

simple hypothesis

names one value of the parameter: $H_0: \theta = \theta_0$

composite hypothesis

allows a range of values: $H_0: \theta \leq \theta_0$ or $H_0: \theta \geq \theta_0$

H_0 always has the equals sign

=, ≤ or ≥ (H_A never does); a two-sided test always uses $H_0: \theta = \theta_0$

test at the boundary

compute z_{obs} or t_{obs} with θ_0, the edge of H_0: the hardest value to reject

Practice: voter turnout

A county election official reports that no more than 30% of the registered voters in Santa Clara County voted in the primary election. A local newspaper suspects turnout was higher, and surveys a random sample of registered voters. (OpenStax, Example 9.1)

the official's figure: no more than 30% voted

(a) What is the parameter, in words?
(b) State H_0 and H_A.
(c) Is H_0 simple or composite? Which tail?

(a) p = the proportion of all registered voters in the county who voted in the primary
(b) H_0: $p \leq 0.30$ H_A: $p > 0.30$
(c) composite; upper-tailed, tested at the boundary $p_0 = 0.30$

Practice: time to graduate

College students are often told a degree takes five years, on average. An advising office believes its students finish faster, and records how long a random sample of recent graduates took to graduate. (OpenStax, Example 9.3)

the usual figure: 5 years, on average

(a) What is the parameter, in words?
(b) State H_0 and H_A.
(c) Is H_0 simple or composite? Which tail?

(a) μ = the mean time to graduate for all of the college's students
(b) H_0: $\mu \geq 5$ H_A: $\mu < 5$
(c) composite; lower-tailed, tested at the boundary $\mu_0 = 5$

Practice: a cure-rate claim

A drug company advertises that its experimental drug cures at least 75% of men with prostate cancer. An independent research group suspects the true cure rate is lower, and follows a random sample of patients who took the drug. (OpenStax, Example 9.8)

the company's claim: a cure rate of at least 75%

(a) What is the parameter, in words?
(b) State H_0 and H_A.
(c) Is H_0 simple or composite? Which tail?

(a) p = the proportion of all men with prostate cancer who are cured by the drug
(b) H_0: $p \geq 0.75$ H_A: $p < 0.75$
(c) composite; lower-tailed, tested at the boundary $p_0 = 0.75$

Example: the conductivity of glass

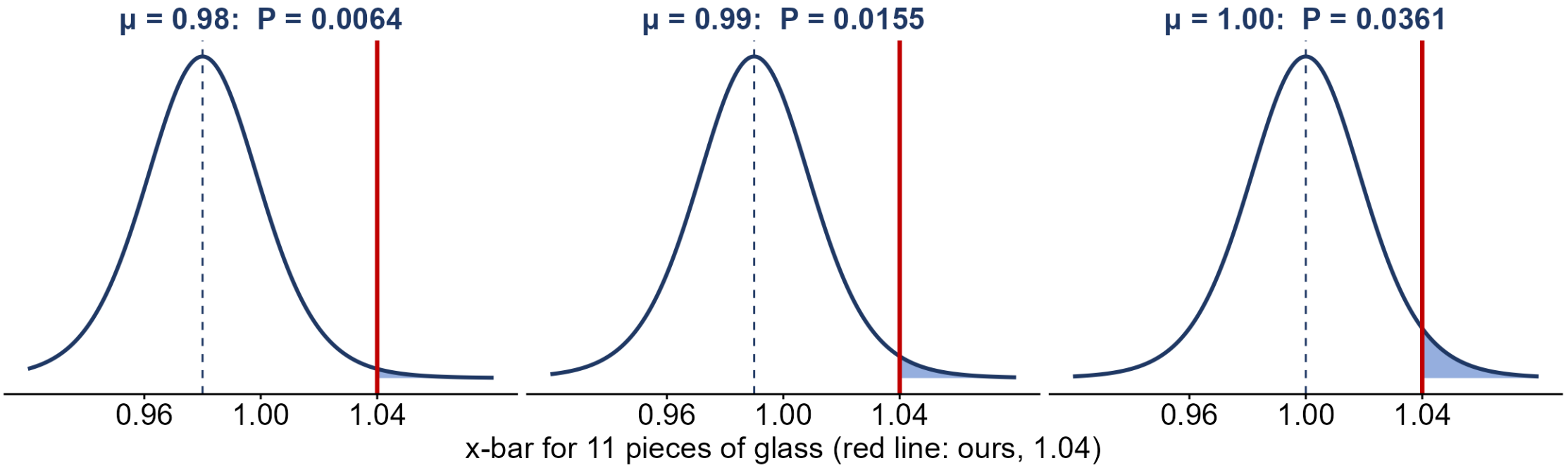
NIST measures the conductivity of 11 randomly selected pieces of a type of glass. Is the mean greater than 1? The population is normal. (OpenStax, Example 9.20)

$n = 11$; $\bar{x} = 1.04$; $s = 0.0659$; $\alpha = 0.05$

(a) What are H_0 and H_A? Which value of μ do we test at?
(b) What are the standard error and t_{obs}?
(c) What is the P-value (use the t-table)? What is the decision?

(a) H_0: $\mu \le 1$ (composite) H_A: $\mu > 1$; test at the boundary, $\mu_0 = 1$
(b) SE = $0.0659 / \sqrt{11} \approx 0.0199$; $t_{obs} = (1.04 - 1) / 0.0199 \approx 2.01$, df = 10
(c) row 10: $1.812 < 2.01 < 2.228$, so $0.025 <$ P-value < 0.05: reject H_0

Why test at the boundary?



Largest P-value at the boundary μ = 1: reject there, and we reject all of H_0

Example: the baker's bread

A baker claims his bread rises more than 15 cm, on average. His customers doubt it, so he bakes 10 loaves. Heights are normal. (OpenStax, Example 9.9)

$n = 10$; $\bar{x} = 17$ cm; $\sigma = 0.5$ cm; $\alpha = 0.05$

(a) What are H_0 and H_A? z or t?
(b) What are the standard error and the test statistic?
(c) What is the P-value? What is the decision?

(a) $H_0: \mu \leq 15$ $H_A: \mu > 15$, test at $\mu_0 = 15$; σ known, so z
(b) $SE = 0.5 / \sqrt{10} \approx 0.1581$; $z_{obs} = (17 - 15) / 0.1581 \approx 12.65$
(c) $P(Z \geq 12.65 \mid \mu = 15) \approx 0 < 0.05$: reject H_0; the loaves average more than 15 cm

Practice: is the mean greater than 12?

A normal population has σ = 1. We want to verify a claim that its mean is greater than 12. (OpenStax, Try It 9.9)

$n = 36$; $\bar{x} = 12.5$; $\sigma = 1$; $\alpha = 0.05$

(a) What are H_0 and H_A?
(b) What are the standard error and z_{obs}?
(c) What is the P-value? What is the decision?

(a) $H_0: \mu \le 12$ $H_A: \mu > 12$, test at $\mu_0 = 12$
(b) $SE = 1 / \sqrt{36} \approx 0.1667$; $z_{obs} = (12.5 - 12) / 0.1667 = 3.00$
(c) $P(Z \ge 3.00 \mid \mu = 12) = 1 - 0.9987 = 0.0013 < 0.05$: reject H_0

Types of errors

Type I error

reject H_0 when it is true: a false alarm; P(Type I error) = α

Type II error

do not reject H_0 when it is false: a missed effect; P(Type II error) = β

power

reject H_0 when it is false: power = 1 – β

α and β

for a fixed n, making one smaller makes the other larger

Decisions and the truth

	H_0 is true	H_0 is false
Reject H_0	**Type I error** false positive probability α	**Correct decision** true positive probability $1 - \beta$ = power
Do not reject H_0	**Correct decision** true negative probability $1 - \alpha$	**Type II error** false negative probability β

Errors in context: climbing equipment

Before a climb, Navah inspects her ropes, harness and anchors. She has used the gear before, so she assumes it is safe, and retires it only if the inspection gives strong evidence that it is not. (OpenStax, Example 9.5)

H_0: Navah's rock climbing equipment is safe

(a) Describe a Type I error in context.
(b) Describe a Type II error in context.
(c) Which error has the greater consequence?

(a) Navah thinks her equipment may not be safe when it really is: she retires good gear
(b) Navah thinks her equipment is safe when it is not: she climbs on it
(c) Type II: unsafe gear can fail during the climb

Errors in context: Genetic Labs

A company called Genetic Labs claims its procedure raises the chance that a pregnancy results in a boy. Statisticians test the claim by comparing the share of boys born to its customers with the usual share. (OpenStax, Example 9.7)

H_0: Genetic Labs has no effect on the baby's sex

(a) Describe a Type I error in context.
(b) Describe a Type II error in context.
(c) Which error has the greater consequence?

(a) we conclude Genetic Labs changes the odds when it has no effect
(b) we conclude it has no effect when it really does change the odds
(c) Type I: people would pay for a procedure that does nothing

Errors in context: a cure-rate claim

A drug company claims its experimental drug cures at least 75% of men with prostate cancer. Patients and doctors use that figure to choose a treatment, and a research group tests the claim. (OpenStax, Example 9.8)

H_0: $p \geq 0.75$ H_A: $p < 0.75$

(a) Describe a Type I error in context.

(b) Describe a Type II error in context.

(c) Which error has the greater consequence?

(a) we conclude the cure rate is under 75% when it is not: patients may skip a drug that works
(b) we accept the 75% claim when the rate is lower: patients rely on a weaker drug
(c) Type II: the 75% claim drives the choice of treatment

Practice: cell phone owners

Marketers believe 92% of U.S. adults own a cell phone. A manufacturer thinks it is lower and surveys 200 adults. (OpenStax, Try It 9.18)

$n = 200$; 174 own a cell phone; $p_0 = 0.92$; $\alpha = 0.05$

(a) What are H_0 and H_A? Is the sample large enough?
(b) What are $\hat{p}$, the standard error and z_{obs}?
(c) What is the P-value? What is the decision?
(d) What are the Type I and Type II errors?

(a) $H_0: p \geq 0.92$ $H_A: p < 0.92$; $np_0 = 184$, $n(1 - p_0) = 16$, both ≥ 15
(b) $\hat{p} = 0.87$; $SE = \sqrt{(0.92 \times 0.08 / 200)} \approx 0.0192$; $z_{obs} \approx -2.60$
(c) $P(Z \leq -2.60 \mid p = 0.92) = 0.0047 < 0.05$: reject H_0
(d) I: say fewer than 92% own one when 92%+ do. II: miss that fewer than 92% do

Statistical power

power

P(reject H_0 | H_0 false) = 1 − β

β

P(do not reject H_0 | H_0 false): the chance of missing the effect

power depends on the truth

it is computed at one assumed true value of the parameter

low power

a real effect is easily missed: "do not reject" is not "H_0 is true"

Example: power for the goggles

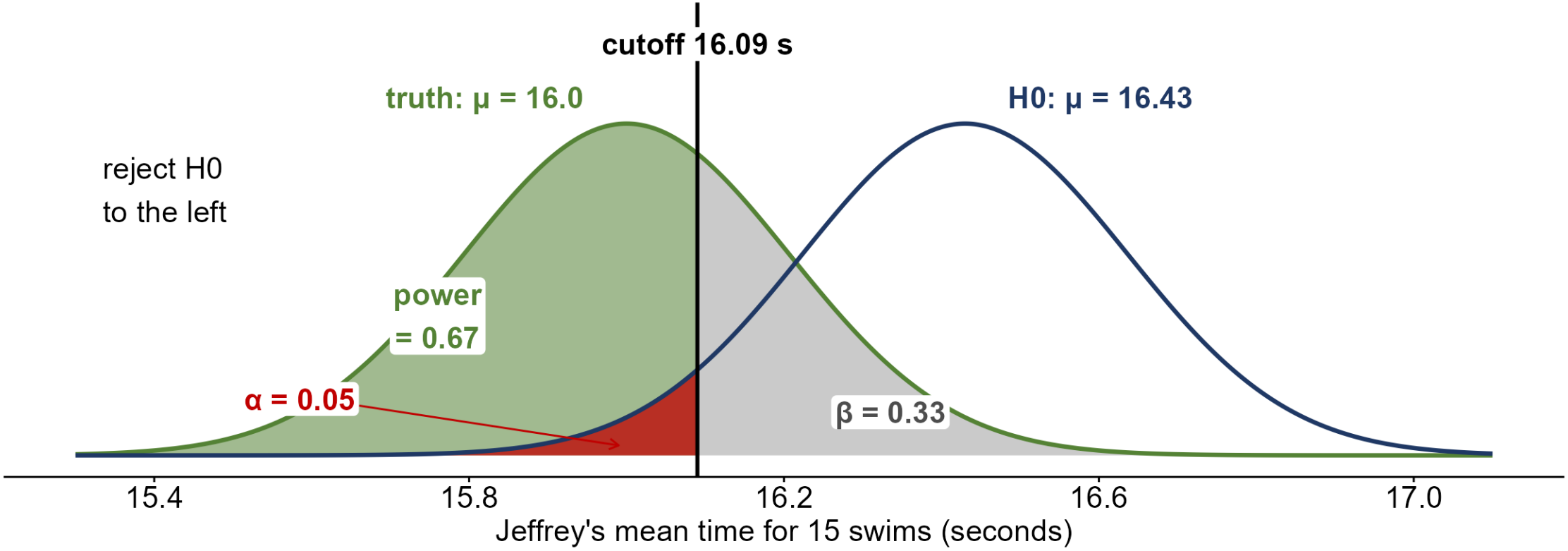
Jeffrey's mean 25-yard freestyle time is 16.43 s, with σ = 0.8. Suppose new goggles really cut his mean to 16.0 s. (OpenStax, Example 9.14)

n = 15; σ = 0.8 s; μ_0 = 16.43; true μ = 16.0; α = 0.05

(a) H_A: μ < 16.43. Below what sample mean does the test reject H_0?
(b) If the true mean is 16.0, what is the power?
(c) What is β? What does it mean?

(a) SE = 0.8 / √15 ≈ 0.2066; reject if $\bar{x} \le 16.43 - 1.645(0.2066) \approx 16.09$
(b) $P(\bar{x} \le 16.09 \mid \mu = 16.0) = P(Z \le (16.09 - 16.0) / 0.2066) = P(Z \le 0.44) = 0.67$
(c) β = 1 − 0.67 = 0.33: a third of the time, the test misses the goggles working

α, β and power


cutoff 16.09 s
truth: μ = 16.0
H0: μ = 16.43
reject H0
to the left
power
= 0.67
α = 0.05
β = 0.33
15.4
15.8
16.2
16.6
17.0
Jeffrey's mean time for 15 swims (seconds)

What raises the power?

more data: n = 30

power 0.67 → 0.90: a smaller SE pulls the two curves apart

a larger $\alpha = 0.10$

power 0.67 → 0.79, but more Type I errors

a bigger effect: $\mu = 15.8$ s

power 0.67 → 0.92: big effects are easier to detect

less variability

a smaller σ shrinks the SE, like a bigger n

Practice: testing water for E. coli

A microbiologist tests a water sample for E. coli before it is approved for drinking. Her test can raise a false alarm or miss bacteria that are there. (OpenStax, Section 9.2, Exercise 19)

H_0: the sample does not contain E. coli

(a) P(she thinks it has E. coli | it does not) = 0.012. Which error, and which letter?
(b) P(she thinks it is clean | it has E. coli) = 0.002. Which error, and which letter?
(c) What is the power of her test? What does it mean?

(a) a Type I error (a false alarm): $\alpha = 0.012$
(b) a Type II error (a miss): $\beta = 0.002$
(c) power = $1 - \beta = 0.998$: she detects E. coli 99.8% of the time it is there

Practice: is the patient sick?

A doctor runs a diagnostic test on a patient. Unless the test gives strong evidence of illness, the doctor treats the patient as healthy and sends them home. (OpenStax, Try It 9.6)

H_0: the patient is not sick

(a) Describe a Type I error in context.
(b) Describe a Type II error in context.
(c) Which error has the greater consequence?

(a) the doctor decides a healthy patient is sick: needless worry and treatment
(b) the doctor decides a sick patient is healthy: the patient goes home untreated
(c) Type II: a sick patient goes untreated